
Forecast system for the turbulence and meteorology at the Paranal Observatory

Performance assessment - monthly report

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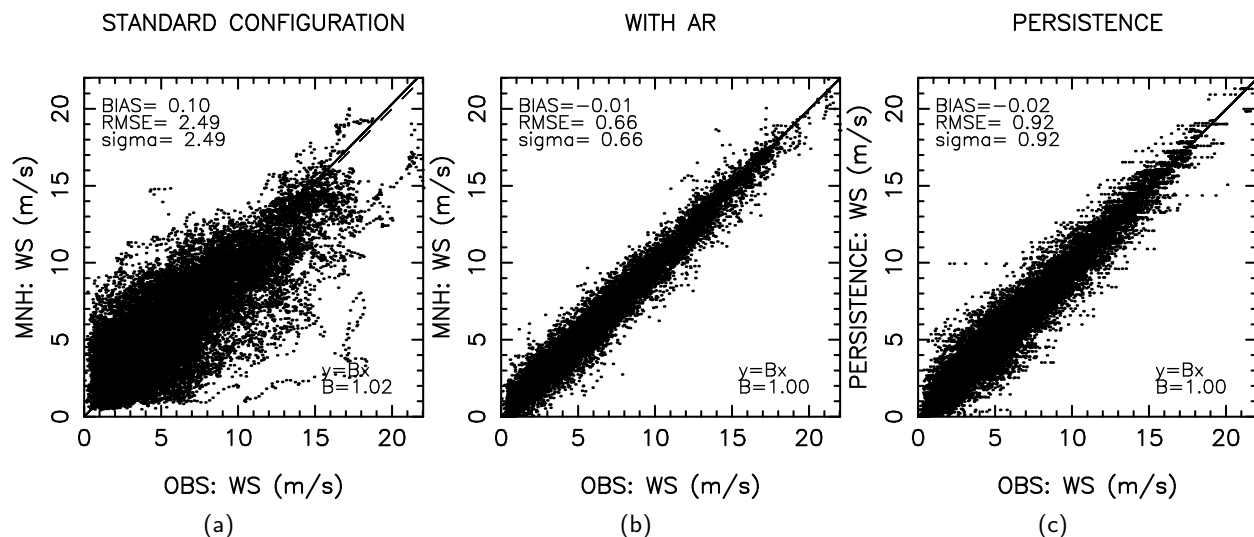


Figure 1: wind speed (m/s): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE.

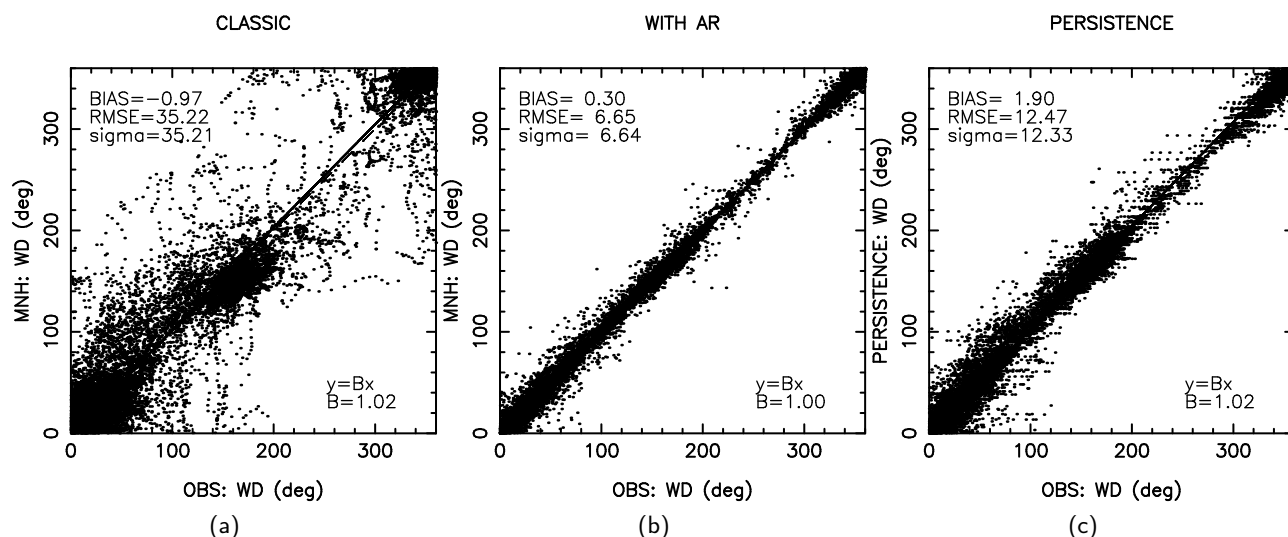


Figure 2: wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR, (c) PERSISTENCE.

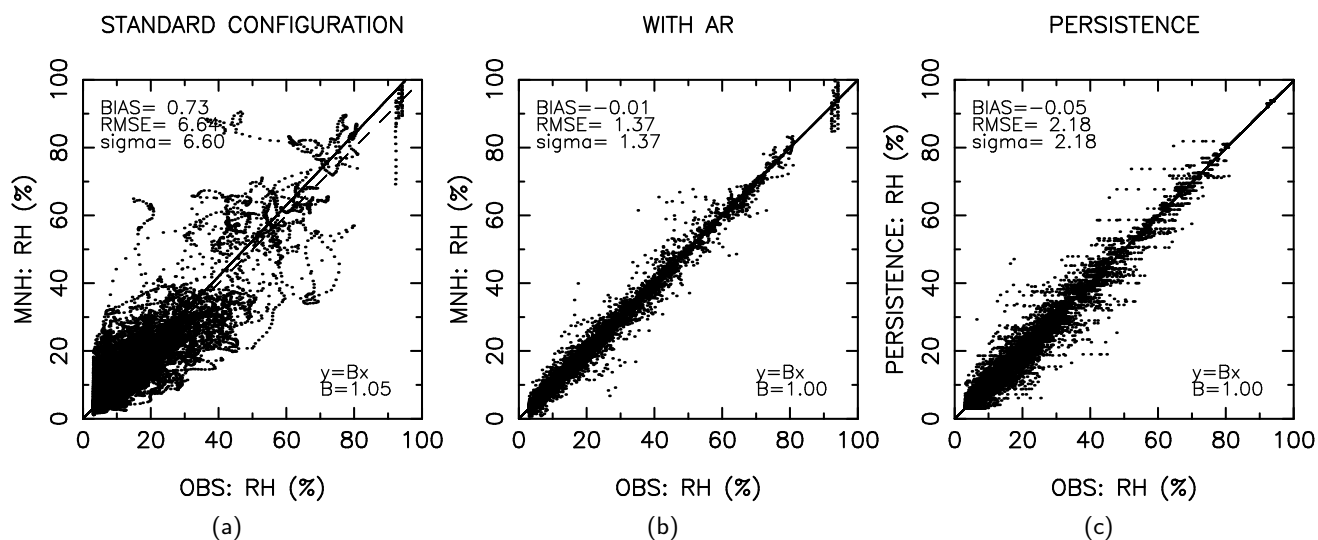


Figure 3: relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE.

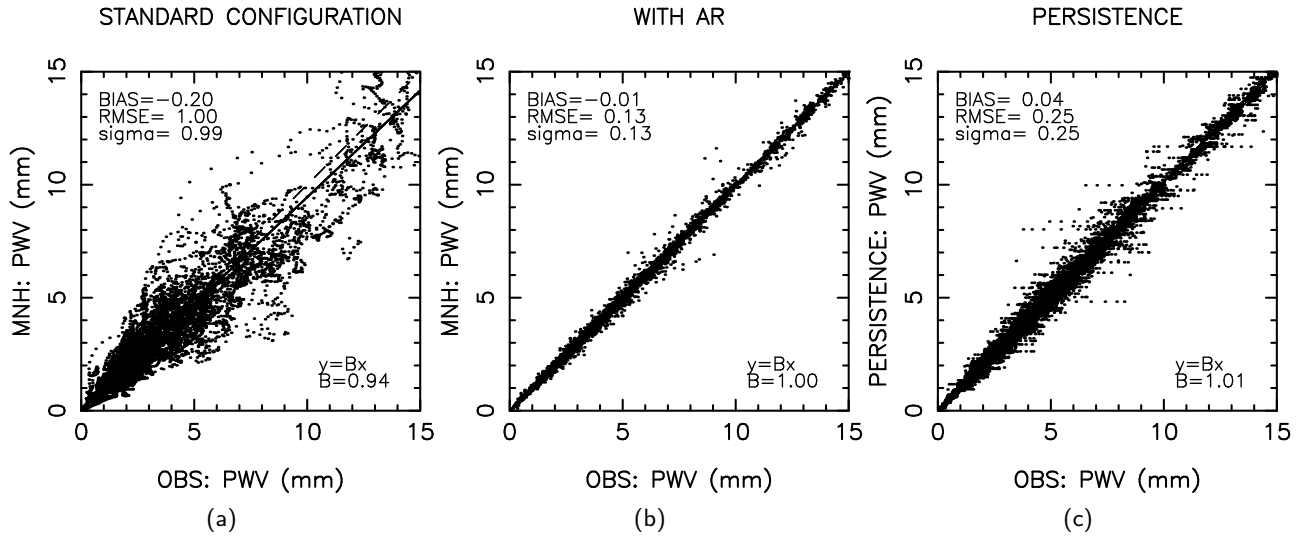


Figure 4: precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE.

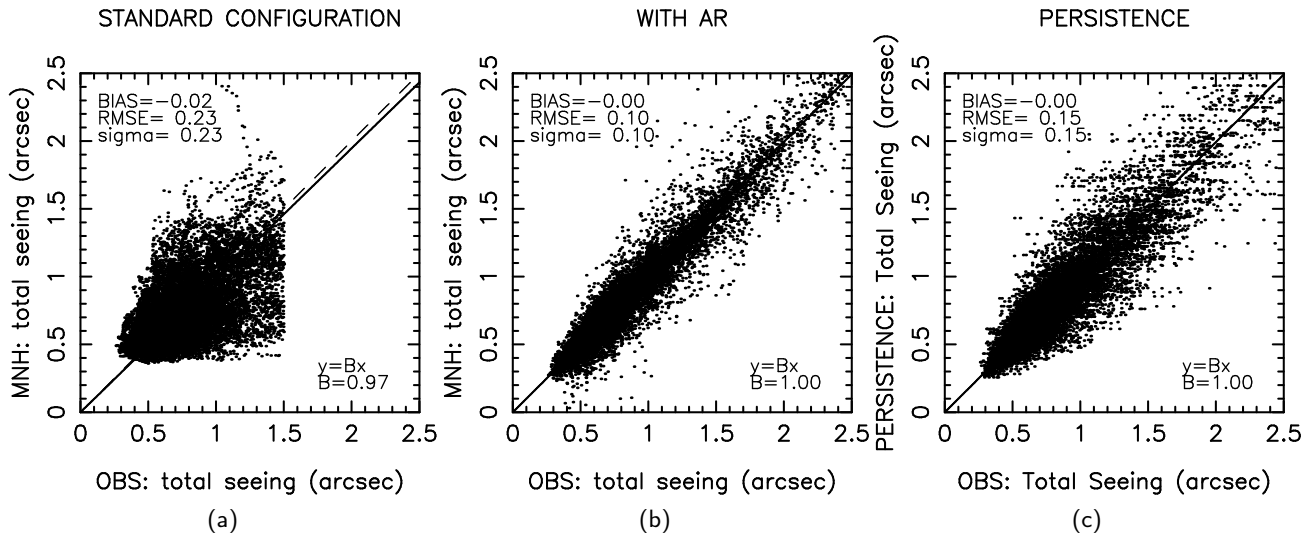


Figure 5: total seeing (arcsec): (a) STANDARD CONFIGURATION ($< 1.5''$), (b) WITH AR (1H), (c) PERSISTENCE.

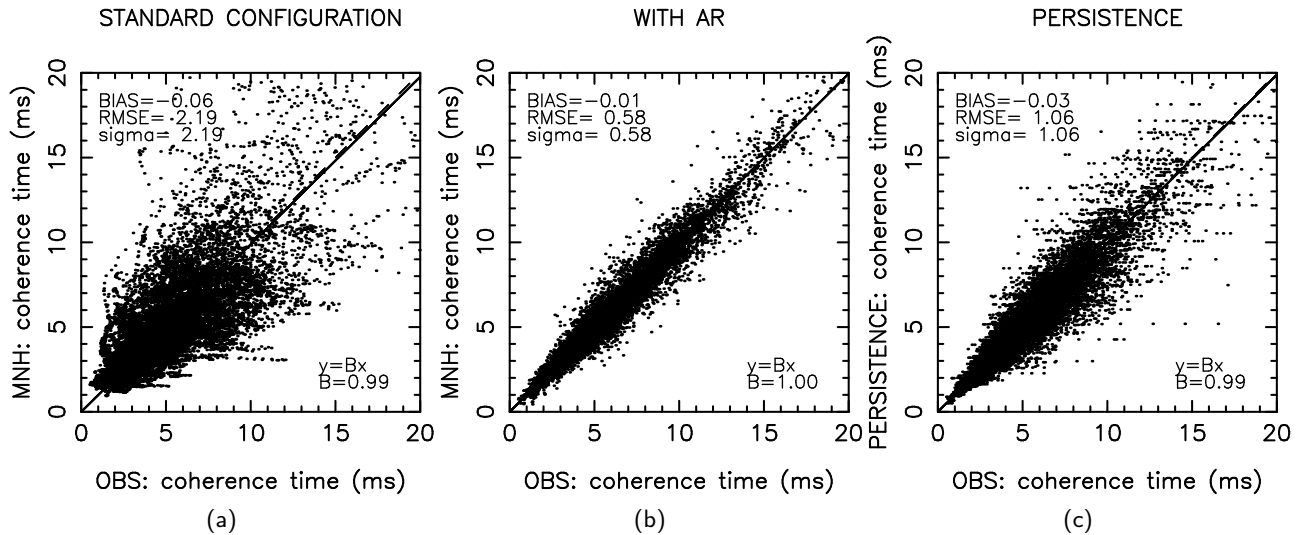


Figure 6: coherence time (ms): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE.

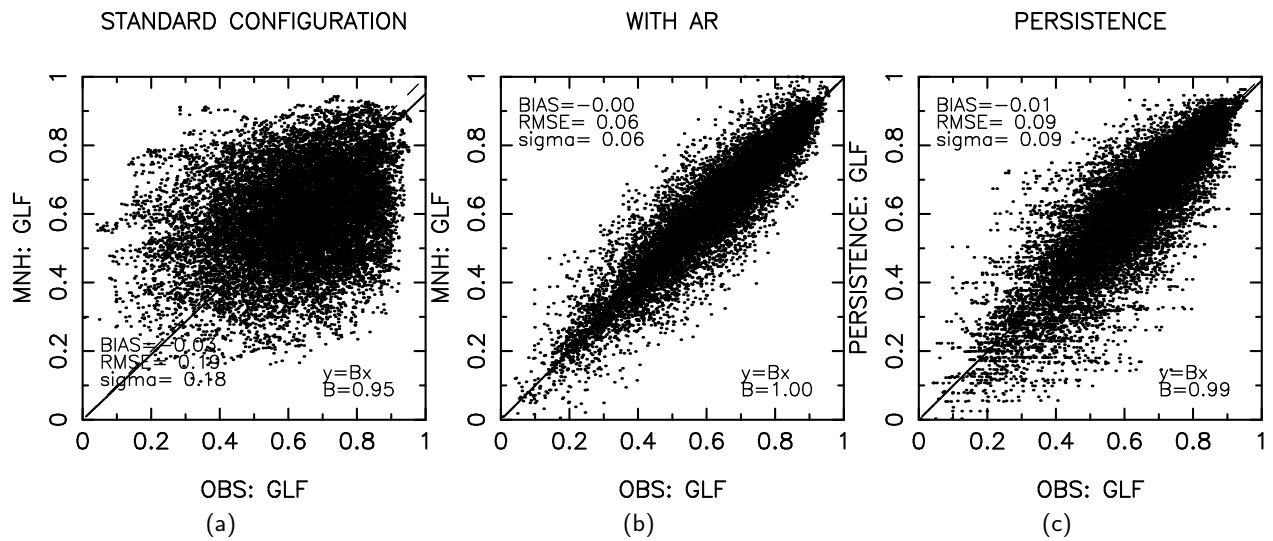


Figure 7: ground layer fraction (dimensionless): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE.

VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS ($\text{m}\cdot\text{s}^{-1}$)	0.10	2.49	2.49	-0.61	2.60	2.53
WD (degree)	-0.97	35.22	35.21	1.64	33.85	33.81
RH (percent)	0.73	6.64	6.60	0.76	6.84	6.79
PWV (mm)	-0.20	1.00	0.99	-0.35	1.00	0.94
SEE (arcsec)	-0.02	0.23	0.23	-0.02	0.22	0.21
TAU (ms)	-0.06	2.19	2.19	-0.63	2.32	2.24
GLF (dimensionless)	-0.03	0.19	0.18	-0.01	0.16	0.16

Table 1: Statistics for variables in standard configuration

VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (persistence)	RMSE (persistence)	SD (persistence)
WS ($\text{m}\cdot\text{s}^{-1}$)	-0.01	0.66	0.66	-0.02	0.92	0.92
WD (degree)	0.30	6.65	6.64	1.90	12.47	12.33
RH (percent)	-0.01	1.37	1.37	-0.05	2.18	2.18
PWV (mm)	-0.01	0.13	0.13	0.04	0.25	0.25
SEE (arcsec)	-0.00	0.10	0.10	-0.00	0.15	0.15
TAU (ms)	-0.01	0.58	0.58	-0.03	1.06	1.06
GLF (dimensionless)	-0.00	0.06	0.06	-0.01	0.09	0.09

Table 2: Statistics for variables processed with AR (1H)

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.5	ws > 7.5
Model data	ws < 4.2	4541	2127	249
	4.2 < ws < 7.5	3034	3970	1347
	ws > 7.5	430	1908	6408
Sample size: 38880; PC=62.1%; EBD=2.8%; POD1=56.7%; POD2=49.6%; POD3=80.1%				

Table 3: Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}		Observations		
		$ws < 4.2$	$4.2 < ws < 7.5$	$ws > 7.5$
Model data	$ws < 4.2$	7377	623	1
	$4.2 < ws < 7.5$	628	6950	386
	$ws > 7.5$	0	432	7617
Sample size: 38880; PC=91.4%; EBD=0.0%; POD1=92.2%; POD2=86.8%; POD3=95.2%				

Table 4: Contingency table for variable wind speed (ms^{-1}) processed with AR (1H)

wind direction		Observations			
degree		North	East	South	West
Model data	North	9401	1028	107	229
	East	1002	1832	560	0
	South	99	508	3567	195
	West	202	89	63	199
Sample size: 38880; PC=78.6%; EBD=1.5%; POD1=87.8%; POD2=53.0%; POD3=83.0%; POD4=31.9%					

Table 5: Contingency table for wind direction (degree) in standard configuration

wind direction		Observations			
degree		North	East	South	West
Model data	North	10453	222	0	36
	East	217	3151	40	0
	South	0	84	4236	29
	West	34	0	21	558
Sample size: 38880; PC=96.4%; EBD=0.0%; POD1=97.7%; POD2=91.1%; POD3=98.6%; POD4=89.6%					

Table 6: Contingency table for wind direction (degree) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	4755	1095	27
	7.1 < rh < 15.3	2696	5193	1357
	rh > 15.3	379	1539	6443
Sample size: 38232; PC=69.8%; EBD=1.7%; POD1=60.7%; POD2=66.3%; POD3=82.3%				

Table 7: Contingency table for variable relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	7408	328	1
	7.1 < rh < 15.3	422	7294	248
	rh > 15.3	0	205	7578
Sample size: 38232; PC=94.9%; EBD=0.0%; POD1=94.6%; POD2=93.2%; POD3=96.8%				

Table 8: Contingency table for variable relative humidity (percent) processed with AR (1H)

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	3402	854	2
	1.9 < pwv < 3.5	443	2596	576
	pwv > 3.5	5	400	3271
Sample size: 18900; PC=80.3%; EBD=0.1%; POD1=88.4%; POD2=67.4%; POD3=85.0%				

Table 9: Contingency table for variable precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	3785	69	0
	1.9 < pwv < 3.5	65	3733	39
	pwv > 3.5	0	48	3810
Sample size: 18900; PC=98.1%; EBD=0.0%; POD1=98.3%; POD2=97.0%; POD3=99.0%				

Table 10: Contingency table for variable precipitable water vapor (mm) processed with AR (1H)

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3678	2340	841
	0.6 < see < 0.8	1828	2382	1886
	see > 0.8	520	1303	3298
Sample size: 37584; PC=51.8%; EBD=7.5%; POD1=61.0%; POD2=39.5%; POD3=54.7%				

Table 11: Contingency table for variable total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	5603	0	597
	0.6 < see < 0.8	0	5649	597
	see > 0.8	423	376	5428
Sample size: 37584; PC=92.3%; EBD=5.6%; POD1=93.0%; POD2=93.8%; POD3=82.0%				

Table 12: Contingency table for variable total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	5707	835	23
	0.6 < see < 0.8	725	4942	546
	see > 0.8	14	669	5877
Sample size: 37584; PC=85.5%; EBD=0.2%; POD1=88.5%; POD2=76.7%; POD3=91.2%				

Table 13: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	6433	11	23
	0.6 < see < 0.8	0	6404	18
	see > 0.8	13	31	6423
Sample size: 37584; PC=99.6%; EBD=0.2%; POD1=99.8%; POD2=99.3%; POD3=99.4%				

Table 14: Contingency table for variable total seeing (arcsec) processed with AR (1H) and accuracy 0.24

coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	4336	1463	107
	$3.4 < \tau < 5.8$	1221	3060	1597
	$\tau > 5.8$	196	1230	4049
Sample size: 37476; PC=66.3%; EBD=1.8%; POD1=75.4%; POD2=53.2%; POD3=70.4%				

Table 15: Contingency table for variable coherence time (ms) in standard configuration and accuracy 0.0

coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	5229	149	639
	$3.4 < \tau < 5.8$	328	4937	635
	$\tau > 5.8$	196	667	5114
Sample size: 37476; PC=88.5%; EBD=4.8%; POD1=90.9%; POD2=85.8%; POD3=80.1%				

Table 16: Contingency table for variable coherence time (ms) in standard configuration and accuracy 1.22

coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	5386	339	2
	$3.4 < \tau < 5.8$	366	5011	400
	$\tau > 5.8$	1	403	5351
Sample size: 37476; PC=91.2%; EBD=0.0%; POD1=93.6%; POD2=87.1%; POD3=93.0%				

Table 17: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 0.0

coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	5745	0	33
	$3.4 < \tau < 5.8$	7	5725	33
	$\tau > 5.8$	1	28	5720
Sample size: 37476; PC=99.6%; EBD=0.2%; POD1=99.9%; POD2=99.5%; POD3=98.9%				

Table 18: Contingency table for variable coherence time (ms) processed with AR (1H) and accuracy 1.22

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	3194	2361	1636
	0.6 < glf < 0.7	1856	2275	2456
	glf > 0.7	707	1121	1665
Sample size: 37476; PC=41.3%; EBD=13.6%; POD1=55.5%; POD2=39.5%; POD3=28.9%				

Table 19: Contingency table for variable ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	4964	566	1758
	0.6 < glf < 0.7	86	5104	1493
	glf > 0.7	707	87	3999
Sample size: 37476; PC=81.4%; EBD=14.3%; POD1=86.2%; POD2=88.7%; POD3=55.2%				

Table 20: Contingency table for variable ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	4991	837	27
	0.6 < glf < 0.7	726	4180	842
	glf > 0.7	40	740	4888
Sample size: 37476; PC=81.4%; EBD=0.4%; POD1=86.7%; POD2=72.6%; POD3=84.9%				

Table 21: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	5707	42	36
	0.6 < glf < 0.7	10	5694	35
	glf > 0.7	40	21	5721
Sample size: 37476; PC=99.1%; EBD=0.4%; POD1=99.1%; POD2=98.9%; POD3=98.8%				

Table 22: Contingency table for variable ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

PARAMETER	STANDARD	WITH AR (1H)
POD1	56.7	92.2
POD2	49.6	86.8
POD3	80.1	95.2
PC	62.1	91.4
EBD	2.8	0.0

Table 23: PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	87.8	97.7
POD2	53.0	91.1
POD3	83.0	98.6
POD4	31.9	89.6
PC	78.6	96.4
EBD	1.5	0.0

Table 24: PODs for wind direction (degree)

PARAMETER	STANDARD	WITH AR (1H)
POD1	60.7	94.6
POD2	66.3	93.2
POD3	82.3	96.8
PC	69.8	94.9
EBD	1.7	0.0

Table 25: PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	88.4	98.3
POD2	67.4	97.0
POD3	85.0	99.0
PC	80.3	98.1
EBD	0.1	0.0

Table 26: PODs for precipitable water vapor (mm)

PARAMETER	STANDARD	WITH AR (1H)
POD1	61.0	88.5
POD2	39.5	76.7
POD3	54.7	91.2
PC	51.8	85.5
EBD	7.5	0.2

Table 27: PODs for total seeing (arcsec) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	93.0	99.8
POD2	93.8	99.3
POD3	82.0	99.4
PC	92.3	99.6
EBD	5.6	0.2

Table 28: PODs for total seeing (arcsec) and accuray 0.24

PARAMETER	STANDARD	WITH AR (1H)
POD1	75.4	93.6
POD2	53.2	87.1
POD3	70.4	93.0
PC	66.3	91.2
EBD	1.8	0.0

Table 29: PODs for coherence time (ms) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	90.9	99.9
POD2	85.8	99.5
POD3	80.1	98.9
PC	88.5	99.6
EBD	4.8	0.2

Table 30: PODs for coherence time (ms) and accuray 1.22

PARAMETER	STANDARD	WITH AR (1H)
POD1	55.5	86.7
POD2	39.5	72.6
POD3	28.9	84.9
PC	41.3	81.4
EBD	13.6	0.4

Table 31: PODs for ground layer fraction (dimensionless) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	86.2	99.1
POD2	88.7	98.9
POD3	55.2	98.8
PC	81.4	99.1
EBD	14.3	0.4

Table 32: PODs for ground layer fraction (dimensionless) and accuray 0.14



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